10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 13 — Sound

Solved Exercise | New Book (PECTAA 2026)

Exercise 13 (Solved)

This exercise covers the production and speed of sound, loudness, pitch and quality, echo and reverberation, acoustics of buildings, and the uses of infrasound and ultrasound.

A. Multiple Choice Questions

13.1 What is required for the production of sound?

(a) A vacuum
(b) Vibrations of a medium
(c) Only air
(d) Light waves

Answer: (b) Vibrations of a medium — sound is produced when the particles of a medium vibrate; without a vibrating medium no sound can be produced or carried.

13.2 An echo is produced due to:

(a) refraction of sound waves
(b) reflection of sound waves
(c) diffraction of sound waves
(d) absorption of sound waves

Answer: (b) reflection of sound waves — the sound striking a distant hard surface bounces back to the listener, and this reflected sound is heard as an echo.

13.3 An example of a longitudinal wave is:

(a) light wave
(b) water wave
(c) sound wave
(d) radiowave

Answer: (c) sound wave — in a sound wave the particles vibrate to and fro along the direction of travel, forming compressions and rarefactions.

13.4 In which medium does sound travel the fastest?

(a) Gases
(b) Liquids
(c) Solids
(d) Vacuum

Answer: (c) Solids — the particles are closest together in a solid, so the disturbance is passed on most quickly; sound cannot travel at all in a vacuum.

13.5 What is the approximate speed of sound in air?

(a) 150 m s^{-1}
(b) 330–350 m s^{-1}
(c) 500 m s^{-1}
(d) 1,000 m s^{-1}

Answer: (b) 330–350 m s^{-1} — at ordinary room temperature the speed of sound in air is about 340 m s^{-1}; it rises slightly with temperature.

13.6 Why is acoustic protection important in building design?

(a) To increase noise pollution
(b) To reduce echo and improve sound quality
(c) To amplify sound
(d) To increase the speed of sound

Answer: (b) To reduce echo and improve sound quality — sound-absorbing surfaces cut down unwanted reflection and reverberation, so speech is heard clearly

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in halls, cinemas and classrooms.

13.7 For a normal person, the audible frequency range for sound waves lies between:

(a) 10 Hz and 10 kHz
(b) 20 Hz and 20 kHz
(c) 25 Hz and 25 kHz
(d) 30 Hz and 30 kHz

Answer: (b) 20 Hz and 20 kHz — a healthy human ear hears sounds from about 20 Hz up to about 20 000 Hz.

B. Short Answer Questions

13.1 How does the medium affect the speed of sound?

The speed of sound depends on the nature of the medium, because sound travels by the vibration of the particles of that medium. The closer the particles are packed and the stronger the force between them, the faster the disturbance is passed on.

So sound is fastest in solids (about 5000 m s^{-1} in steel), slower in liquids (about 1500 m s^{-1} in water) and slowest in gases (about 340 m s^{-1} in air). In a vacuum there are no particles, so sound cannot travel at all. In a gas the speed also increases with temperature and with humidity, because the particles move faster and pass the disturbance on more quickly.

13.2 Does sound waves show reflection?

Yes, sound waves show reflection. When they strike a hard, smooth surface such as a wall, a cliff, or the ceiling of a hall, they bounce back into the same medium just as light does. The angle of incidence is equal to the angle of reflection.

This reflected sound is called an echo. If the reflected sound reaches our ear more than 0.1 s after the original sound, the ear can hear it separately as an echo; if it arrives sooner, it merges with the original sound and makes it louder, which is what happens in a small room. Sound can be reflected from a curved surface as well — a large curved ceiling or wall can focus the sound to one point, and this property is used in some halls and in old-style whispering galleries.

13.3 What factors influence the loudness of sound?

Loudness depends mainly on the amplitude of the sound wave. A bigger amplitude means the particles vibrate through a larger distance, which carries more energy and produces a louder sound. Loudness is measured in decibels (dB).

Other factors also matter: the distance from the source, because the intensity falls as the sound spreads out; the surface area of the vibrating source, which is why a tuning fork kept on a wooden table sounds louder than one held in air; the density and nature of the medium; and the sensitivity of the listener's ear. A forced vibration of a larger body — the sound box of a guitar or a drum membrane — also adds to the loudness.

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13.4 Why do we hear an echo when we clap near a building?

When we clap near a large building, the sound waves travel to the wall, strike the hard smooth surface and are reflected back to us. If the wall is far enough, this reflected sound reaches our ear after a short delay and we hear it as a separate clap — an echo.

For the echo to be heard separately, the reflected sound must arrive at least 0.1 s after the original sound, because the ear cannot separate two sounds that are closer in time than that. As sound travels 340 m in one second, the total distance travelled must be at least 34 m, so the reflecting surface must be at least about 17 m away. A nearby wall reflects the sound too soon, so the echo merges with the clap and only makes it louder.

13.5 What is ultrasound, how is it used in medicine?

Ultrasound is sound of frequency higher than 20 kHz, that is, above the range which a human ear can hear. It is produced by a piezoelectric transducer, which converts electrical vibrations into very fast mechanical vibrations of a crystal.

In medicine ultrasound is used in several ways. In sonography (ultrasound scanning) high frequency waves are sent into the body and the echoes from different tissues are converted into an image on a screen; this is how an unborn baby is seen and how the heart, liver, kidneys and other organs are examined. It is safe, painless and needs no X-rays. Ultrasound is also used to break kidney stones into tiny pieces, to clean teeth in dentistry, to measure the flow of blood, and in physiotherapy to warm deep tissues. Industrially it is used to detect cracks in metals and to measure the depth of the sea (sonar).

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C. Constructed Response Questions

13.1 Why does a tuning fork placed on a wooden table produce a louder sound than when held in air? What does this tell us about the role of surface area in sound propagation?

A tuning fork has only two small prongs, so the area of it that touches the air is very small. It sets only a thin layer of air vibrating, and most of its energy is used in vibrating the fork itself, so the sound produced when it is held in air is faint.

When the stem of the fork is pressed on a wooden table, the fork forces the whole table top to vibrate with it. The table has a very large surface area, so it sets a large volume of air vibrating and pushes the air much more effectively. A bigger area of air is disturbed, so more energy is radiated as sound and we hear a louder note. The table is said to act as a sounding board, and it also acts like a forced vibration system driven by the fork.

This shows that the loudness of a sound depends on the surface area of the vibrating source: a larger vibrating surface pushes more air and produces a louder sound. That is why musical instruments have sound boxes, why loudspeakers have large cones or baffles, and why the thin membrane of a drum produces a much louder sound than a vibrating comb.

13.2 If sound needs a medium to travel, how does the bell jar experiment prove this requirement? What happens when air is removed, and why?

An electric bell is suspended inside a glass bell jar from which air can be pumped out. The jar is placed on a rubber base plate and the pump connection is closed. When the bell is switched on with the air still inside, its ringing is heard clearly through the glass.

Now the air is slowly pumped out of the jar. The ringing becomes fainter and fainter, although the hammer of the bell can still be seen striking the gong. When a good vacuum has been produced, no sound at all is heard, even though the bell is visibly vibrating.

The reason is that sound needs a material medium to carry its energy: with the air removed there are almost no particles to be compressed and passed on, so the vibrations of the bell cannot reach the outside. When air is let back into the jar, the ringing is heard again and becomes louder as the pressure returns. The experiment therefore proves that sound cannot travel through a vacuum and needs a medium such as air, water or a solid.

13.3 A person hears his echo when shouting near a tall building. Why must the building be at least 17 metres away for the echo to be heard clearly?

The human ear can distinguish a reflected sound from the original sound only if the two reach the ear at least 0.1 s apart. This is called the persistence of hearing.

Sound travels about 340 m in one second, so in 0.1 s it covers $s = v\,t = 340 \times 0.1 = 34$ m. This distance is the total path travelled by the sound — from the person to the building and back again to the person, which is twice the distance to the building. Therefore the building must be at least $34 / 2 = 17$ m away.

If the building is nearer than 17 m, the echo returns in less than 0.1 s and the ear cannot separate it from the shout; the two sounds simply mix and make the shout seem louder. This is why an echo is heard in an open field near a distant hill or a large building, but not in a small room.

13.4 Why do voices of men sound deeper than those of women or children, even when they are speaking at the same intensity? Explain using the relationship between frequency and pitch.

The pitch of a sound depends on its frequency: the higher the frequency, the higher (sharper) the pitch of the sound; the lower the frequency, the deeper (flatter) the sound. Intensity or loudness depends on amplitude, which is a separate property.

The vocal cords of an adult man are longer and thicker (about 17–25 mm) than those of a woman or a child (about 13–17 mm). Thicker and longer cords vibrate more slowly, so they produce a lower frequency — typically about 100–130 Hz for men and about 200–260 Hz for women, and even higher for children.

Because the frequency is lower, the pitch is lower, so a man's voice sounds deeper. If the loudness of the two voices is the same (same intensity), we still hear the difference clearly, because pitch and loudness are independent qualities of sound.

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13.5 In what ways do soft materials like carpets or curtains improve the acoustics of a room? How do they affect echoes and sound quality?

Soft, porous materials such as carpets, curtains, cushions, acoustic tiles and wooden panelling absorb sound energy instead of reflecting it. The sound waves enter the tiny pores of the material, the air inside them is made to vibrate and the energy is converted into a very small amount of heat. Only a little sound is reflected back.

Because less sound is reflected, the unwanted echo and reverberation (the prolonged mixing of reflected sounds) are reduced. The result is that a speaker's words are heard clearly without blurring, background noise is reduced, and the sound does not become painfully loud. In this way the room becomes acoustically good.

A bare hall with plaster walls and a stone floor gives strong, long-lived reflections, so speech sounds confused and noisy. Lining it with carpets and curtains removes the harshness. In a recording studio or a cinema the walls are covered with soft, wedge-shaped absorbers so that the sound can be controlled exactly, and in a library the soft furnishings keep it quiet.

13.6 When a person claps near a canyon and hears the echo after a few seconds, how can they calculate the distance to the canyon? What principle is used in this method?

The principle used is the reflection of sound, together with a known speed of sound. The clap travels from the person to the canyon wall and then back as an echo, so the total distance covered is twice the distance to the wall.

Method: measure the time t (in seconds) between the clap and the echo with a stopwatch or a slow-motion video. Take the speed of sound in air as v = 340 m s^{-1} (or 343 m s^{-1} on a cool day). Then use s = vt for the total path, so the distance to the canyon is d = v t / 2.

Example: if the echo is heard after 3 s, then the sound travelled s = 340 × 3 = 1020 m in all, so the distance to the canyon is 1020 / 2 = 510 m. The same method is used by ships with sonar to measure the depth of the sea and by bats and dolphins to find their way — they send out a sound and judge the distance from the time taken by the echo.

D. Comprehensive Questions

13.1 What is the approximate speed of sound in air, and what factors affect it?

The speed of sound in air at ordinary room temperature (about 20 °C) is approximately 340 m s^{-1}, or about 330–350 m s^{-1} depending on the conditions. For rough calculations the value 340 m s^{-1} is used.

The factors that affect the speed of sound in air are:

1. Temperature: the speed increases by about 0.6 m s^{-1} for every 1 °C rise. The particles of warm air move faster, so they pass the disturbance on more quickly. At 0 °C the speed is about 331 m s^{-1} and at 30 °C it is about 349 m s^{-1}.

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2. Humidity: moist air is less dense than dry air, so sound travels slightly faster in humid air.
3. Pressure: at constant temperature a change of atmospheric pressure does not change the speed much, because the density changes in the same proportion.
4. Nature of the medium: speed is much greater in liquids and solids — about 1500 m s^{-1} in water and about 5000 m s^{-1} in steel — and it falls to zero in a vacuum.
5. Wind: a wind blowing in the direction of the sound increases its speed relative to the ground, and a wind blowing against it decreases the speed.

13.2 Why does sound travel faster in solids than in liquids and gases? Explain.

Sound is a mechanical wave: it travels because each particle pushes or pulls the next one and so passes the disturbance on. The speed of this transfer depends on two things — how close the particles are to one another and how strongly they are bound together.

In a solid the particles are packed very close together and are held by strong intermolecular (elastic) forces, so a push given to one particle is passed almost instantly to its neighbour. The solid also has a very large bulk and shear modulus of elasticity, which means it resists compression strongly and returns energy quickly; both of these make sound travel fastest — for example about 5000 m s^{-1} in steel.

In a liquid the particles are farther apart and the forces are weaker, so the disturbance is passed on more slowly (about 1500 m s^{-1} in water). In a gas the particles are very far apart with almost no binding force, so the transfer is slowest — about 340 m s^{-1} in air, and slower still in a thin gas. In a vacuum there are no particles at all, hence sound cannot travel.

13.3 How do amplitude and frequency, affect the quality of sound? Explain.

Amplitude controls loudness. The energy carried by a sound wave is proportional to the square of its amplitude, so a larger amplitude gives a louder sound and a smaller amplitude a softer one. Loudness is measured in decibels; the amplitude of the vibration depends on how hard the source is struck or blown.

Frequency controls pitch. A high frequency produces a high, shrill note and a low frequency a deep, flat note. A normal ear hears frequencies between 20 Hz and 20 kHz; below 20 Hz is infrasound and above 20 kHz is ultrasound, and neither is heard by us.

Quality (timbre) is a third, separate property. It is decided by the waveform of the sound, that is, by the number and relative strength of the overtones (harmonics) mixed with the fundamental frequency. This is why a piano and a flute playing the same note with the same loudness are still recognised as different instruments: their amplitudes and frequencies may match, but their mixtures of overtones are different. So amplitude affects loudness, frequency affects pitch, and the pattern of overtones affects quality.

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13.4 How does echo occur? Explain.

An echo is the repetition of a sound heard after the original sound, due to the reflection of sound waves from a distant hard surface such as a wall, a cliff, a hill or a large building.

When a sound is produced, its waves travel outward. If they meet a large, hard, smooth surface, most of the energy is reflected back and travels towards the listener. If the surface is at least about 17 m away, the reflected sound reaches the ear more than 0.1 s after the direct sound. The ear then hears it as a separate sound and we call it an echo.

Conditions for an echo: the reflecting surface must be large, hard and smooth — soft surfaces such as curtains absorb the sound and give no echo; and the surface must be far enough away (17 m or more). An echo is useful in sonar and depth sounding, and in ultrasound scanning, but in a hall it spoils the sound, which is why soft materials are used to absorb the reflected sound.

13.5 How do reflection, refraction, and diffraction affect the behaviour of sound waves?

Reflection. Sound bounces back from hard surfaces, obeying the law that the angle of incidence equals the angle of reflection. This gives echo and reverberation. A curved surface can focus or spread the reflected sound; a concave ceiling focuses it towards the seats.

Refraction. When sound passes from one medium into another, or through layers of air at different temperatures, its speed changes and its path bends. On a hot day the air near the ground is warmer, so sound travelling from a distant source bends upward and the sound seems to fade; over a cold lake the sound bends downward and carries much farther. Refraction also occurs when sound enters water from air.

Diffraction. Sound spreads out after passing through a doorway or around the edge of an obstacle, because its wavelength (about 1 m for ordinary speech) is comparable with the size of the gap. This is why we hear a person talking in the next room, or round a corner, without seeing him. Low-frequency (long wavelength) sounds diffract more than high-pitched ones, so the bass of a band is heard from farther away.

In short, reflection returns the sound to the listener, refraction bends its path, and diffraction bends it round obstacles and spreads it out.

13.6 What are infrasound and ultrasound and how are they used in different applications?

Infrasound is sound of frequency below 20 Hz, which the human ear cannot hear. It is produced by very large or slow vibrations — earthquakes, volcanic eruptions, thunder, ocean waves, wind over mountains, and large machines. It travels a long way with very little absorption. Uses: seismographs and monitoring stations detect earthquakes and volcanic activity, underground nuclear tests are identified, infrasound from storms is used in weather forecasting, and some animals such as elephants and whales use it to communicate over long distances.

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Ultrasound is sound of frequency above 20 kHz, also inaudible to man. It is produced by a piezoelectric crystal and can be focused into a narrow beam. Uses: sonar and echo sounding to measure the depth of the sea and to locate shoals of fish and submarines; medical diagnosis (sonography) to see an unborn baby and internal organs; lithotripsy to break kidney stones; dentistry and surgical cleaning; cleaning delicate parts in a liquid bath; non-destructive testing to find cracks and holes inside metal castings; and industry uses it to weld plastics, drill hard materials and to make ultrasonic distance sensors.

Infrasound is used where the wavelength is huge and the sound must travel very far, while ultrasound is used where a short wavelength and a narrow beam are needed to get detailed information about small objects.

E. Numerical Problems

13.1 At a particular temperature, the speed of sound in air is 333 m s^{-1}. If the wavelength of a sound wave is 3 cm, calculate the frequency of the sound wave. Is this frequency in the audible range of the human ear?

Given: v = 333 m s^{-1}
λ = 3 cm = 0.03 m

To find: f = ?

Formula: f = v / λ

Solution:
f = 333 / 0.03
f = 11 100 Hz
f = 11.1 kHz

Answer: f = 11.1 kHz — yes, it lies between 20 Hz and 20 kHz, so it is audible

13.2 A clock chimes 48 times in 1 minute. Calculate the frequency and period of the chimes.

Given: Number of chimes = 48
Time = 1 minute = 60 s

To find: f = ?, T = ?

Formula: f = N / t, T = 1 / f

Solution:
f = 48 / 60
f = 0.8 Hz
T = 1 / 0.8
T = 1.25 s

Answer: f = 0.8 Hz, T = 1.25 s

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13.3 A car horn emits a sound with a wavelength of 0.5 m. If the speed of sound in air is 340 m s^{-1}, calculate the frequency of the sound produced by the car's horn.

Given: λ = 0.5 m
v = 340 m s^{-1}

To find: f = ?

Formula: f = v / λ

Solution: f = 340 / 0.5
f = 680 Hz

Answer: f = 680 Hz

13.4 A wave travels with a speed of 500 m s^{-1} and has a wavelength of 2 metres. Calculate the time period of the wave.

Given: v = 500 m s^{-1}
λ = 2 m

To find: T = ?

Formula: f = v / λ, T = 1 / f

Solution: f = 500 / 2 = 250 Hz
T = 1 / 250
T = 0.004 s

Answer: T = 0.004 s (4 ms)

13.5 A research boat sends a sound wave straight to the seabed and receives the echo 2 seconds later. The speed of sound in seawater is 1600 m s^{-1}. Find the depth of the sea at this position.

Given: v = 1600 m s^{-1}
t = 2 s
the sound goes down and comes back, so the distance is 2d

To find: d = ?

Formula: 2d = v t ⇒ d = v t / 2

Solution: d = (1600 × 2) / 2
d = 3200 / 2
d = 1600 m

Answer: d = 1600 m

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13.6 A person claps his hands near a mountain and hears the echo after 6 seconds. If the speed of sound is 343 m s^{-1}, what is the distance of the mountain from the person?

Given: v = 343 m s^{-1}
t = 6 s

To find: d = ?

Formula: 2d = v t ⇒ d = v t / 2

Solution: d = (343 × 6) / 2
d = 2058 / 2
d = 1029 m

Answer: d = 1029 m

Points to Remember

- Sound is a longitudinal mechanical wave — it needs a medium.
- Speed: solids > liquids > gases; in air ≈ 340 m s^{-1} at 20 °C.
- Loudness ∝ amplitude (dB); pitch ∝ frequency; quality ∝ waveform/overtones.
- Audible range 20 Hz – 20 kHz; below is infrasound, above is ultrasound.
- Echo: 2d = v t, and the reflector must be at least 17 m away.

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